

AI / Engineering Dossier

Student Success Early-Signal & Academic Planning System

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Purpose and AI Use

I used generative AI as a support tool throughout this project. It helped me plan the work, think through different implementation options, review and revise parts of the code, troubleshoot problems, and organize the final report and presentation. I did not treat AI output as automatically correct. I ran the code myself, checked the results, reviewed the project files, and made the final decisions about what to keep or change.

The main AI tools I used were ChatGPT and Codex. ChatGPT was used more broadly for planning, coding support, debugging, and writing. Codex was used near the end of the project to build and polish the dashboard demo after the required backend work was already complete.

Significant AI Assistance

Project Area	Tool	How AI Helped	What I Did
Project planning	ChatGPT	Helped break the project instructions into smaller steps.	I chose the scope and decided to focus on a simple design that I could explain and test.
Backend development (coding help)	ChatGPT	Helped suggest code structure and revise parts of the code I wanted to change.	I ran the code locally, reviewed the outputs, checked whether the results made sense, and decided which changes to keep before committing them.
Debugging and results	ChatGPT	Helped explain errors and compare model results.	I used the actual results from my local runs as the source of truth. I also kept the model and threshold locked after the final test evaluation.
Demo dashboard (coding help)	Codex	Built and polished the Streamlit dashboard presentation layer after the required backend was finished.	I worked on a separate demo-polish branch, reviewed the changes, ran the app, checked each screen, and decided what should stay in the final demo.
Report and presentation	ChatGPT	Helped organize the report and improve wording after I wrote the first draft	I read through and revised the report in my own wording, corrected details, and made the final choices about what to disclude.

How I Verified AI-Assisted Work

AI suggestions were checked through direct execution and review instead of being accepted as-is. The most important verification steps are listed below.

Local runs: I ran the main project commands myself and used the outputs from my own environment for the final results.

Training and test separation: I confirmed that the held-out test set was not used to choose the final model or tune the threshold, and there was no student overlap between training and test data.

Metric checks: I reviewed the saved precision, recall, F1, confusion counts, fairness results, and failure cases before using them in the report or demo.

Planner validation: Every required advisor-case plan was checked by a separate validator. All 12 plans passed.

Automated tests: The project includes tests for restricted model features and planning rules such as prerequisites, completed courses, and credit limits.

Git review: For the Streamlit work, I reviewed the demo-polish branch against the main branch to confirm that the presentation changes did not change the required backend or final results.

Manual review: I opened the final dashboard and checked the advisor cases, fairness page, failure analysis, decision log, and reproducibility information before considering the demo finished.